Homework — 1.2.4 Types of Programming Language — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] A programming paradigm is a style/approach to writing programs (e.g. procedural, object-oriented) (1). No single paradigm is best for every problem because different problems are easier to express, maintain or reuse in different paradigms, so the programmer chooses the one that fits the task (1).

2. [3] Any three (1 each): uses sequence, selection and iteration; built from procedures/subroutines (functions); statements executed in a defined order; specifies *how* the task is done step by step; uses variables to hold state. Max 3.

3. (a) **[2]** It reads two numbers; it subtracts the first from the second (second − first) (1) and outputs the result. Because of the `BRP`, it always outputs the result regardless of sign here — accept "outputs the second number minus the first number" (1). *(Both OUT paths output ACC, so the difference is always shown.)*

(b) **[2]** `BRP` branches to the given address only if the Accumulator is positive (≥ 0) (1). It is used to test the sign of the subtraction result / to direct the flow of the program depending on whether the result is negative or not (1).

4. [3] (a) Immediate = **4** (operand is the value itself) (1). (b) Direct = **7** (value held at address 4) (1). (c) Indirect = **2** (address 4 holds 7, so go to address 7, whose value is 2) (1). (d) Effective address for indexed = base 4 + index 3 = **address 7** (not separately marked; reward in feedback — the 3 marks are for (a)–(c)).

5. [2] Inheritance = a subclass acquires (inherits) the attributes and methods of a superclass, and can add to or override them (1). Benefit (1): reduces duplicated code / promotes reuse (DRY), so the program is easier to maintain and extend. Max 2.

Part B (6 marks)

6. [3] Advantages (1 each, max 2): simple / easy to manage and plan; well documented; clear structure with defined milestones/stages. Disadvantage (1): inflexible / hard to go back and change a completed stage; the client only sees the finished product at the end, so requirement errors are costly. Max 3.

7. [2] Static linking copies the library code into the executable, making it larger but self-contained (1). Dynamic linking keeps the library as a separate file (e.g. .dll) linked at run time, making the executable smaller and the library shareable/updatable (1).

8. [1] The Accumulator temporarily holds the result of arithmetic and logic operations performed by the ALU (1).

Total: 14 + 6 = 20